ImagoEngine Annual Impact Plan

Innovationships | Bachelor of Innovation UCCS

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Dates Covered: May 2026 - May 2027

Executive Summary

Goal	Description	Key Deliverables
Engine Core	Build a professional ECS-based game engine with an organized and well documented code base.	• ECS engine API • Physics engine • Rendering pipeline • Catch2 unit test
Editor Foundation	Mockup and build a foundation for an intuitive editor for the game engine.	• Research existing game engine editors • Wireframe mockups • Basic integration in project with Dear ImGUI
Documentation	Create understandable and navigable documentation for the backend and “how to use” resources.	• Generated Doxygen backend documentation • Front end documentation on website • Video/article basic learning resources
Business Progress	Establish SD Creations as a professional business with ImagoEngine as its flagship product.	• Business bank account • Official LLC formation • Written IP and licensing research
Social Media Impact	Build an audience around ImagoEngine to gain potential users.	• Website has About Us, Documentation, and Dev Logs tabs • Social Media post per phase

Vision and Mission

ImagoEngine is a professional-quality, commercially released 2D game engine built from scratch in C++ that acts as the foundational technology and flagship product of

SD Creations. Its name comes from the Latin *Imago Dei*, the Image of God, named after the conviction that every person carries a reflection of their Creator that ought to be expressed through the act of creating. ImagoEngine serves two purposes: it is the platform SD Creations will use to build and publish its own 2D game titles, and it is a tool the will enable others to create. ImagoEngine aims to revolutionize 2D game development by making a shift towards Data Oriented Design in game programming while still giving designers intuitive access to the game's systems, entities, and components within the editor.

Goals

Goal 1: Build a professional, modular engine core

Develop foundational systems of ImagoEngine including the ImagoECS core library, physics engine, and asset rendering system. This will follow a strict Entity-Component-System core, and will take the form of a disciplined, professional codebase that passes rigorous unit tests.

Goal 2: Build an intuitive, user-friendly editor application

Research current popular game engine editors and apply their design concepts to and ECS based engine. Create an ImagoEditor mock-up that makes development as intuitive as possible for designers and artists who may not be as comfortable with ECS as programmers. Build the foundation of this editor mock-up using Dear ImGui and conduct rigorous unit tests.

Goal 3: Create documentation and learning resources for ImagoEngine

Generate Doxygen documentation from Doxygen comments throughout code and use it as a base to create backend documentation for how ImagoECS core systems work. Then build fully human-written, intuitive and descriptive front-end documentation for how the user interacts with the code base and accomplishes the most common tasks with ImagoECS game development. Create learning resources for these most common tasks and editor usage in many several formats including web articles and videos.

Goal 4: Establish SD Creations as a legitimate business

Create domain and website to publish the created documentation and learning resources. Research and pursue intellectual property endeavors to protect ImagoEngine as a product while still keeping it open source. Start moving SD Creations from personal project to a formally structured small business by opening a business bank account and forming an LLC.

Goal 5: Build an audience and community around the project

Publicly share the development journey in order to build a social media presence, get developers excited about ECS, inspire future developers to create, and build a base of future users for ImagoEngine.

Activities

Summer 2026: Design, Planning, and Learning

- Study and test core technologies (SFML, Dear ImGui, SDL2, nlohmann/json, spdlog)
- Study ECS theory and current ECS technologies.
- Study 2D physics theory.
- Define the engine’s architecture, technology stack, and library strategy
- Collect this information into a technical design document.

Fall 2026: Engine Core and Physics Development

- Build ECS core: entity management, spares-set component pools, command buffering, event broadcasting, system scheduling, and scene management.
- Design, test, and implement a self-built physics and collision system using broad phase and narrow phase detection, collision resolution, and rigid body integration.
- Catch2 unit tests throughout and rigorously at the end of development.
- Create basic website and a template for dev logs for social media.
- Start posting on social media with an intent to funnel viewers to the website.
- LLC and intellectual property research

Spring 2027: Asset Renderer and Editor Foundation

- Complete the renderer and asset system
- Create/compile documentation for ImagoECS libary, Physics Engine, and Asset Renderer.
- Research current game engine editor designs and mock-up ImagoEditor's design, answering all the question that come with integrating an editor with ECS.
- Build the foundation of the visual editor using Dear ImGui.
- Start production of tutorial content for the editor and C++ code integration.
- Form the LLC, business account, and website domain.
- Pursue desired licensing/intellectual property efforts revealed in previous research,

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Expected Outcomes

Goal	Expected Outcome	How It's Measured
1. Engine	ECS core, physics and collision, and asset renderer system completed, tested, and tagged.	Version-tagged milestones in the repository. Automated unit tests passing. Phase completion against the project timeline
2. Editor	Editor designed, mocked-up and answering major ECS designer questions with foundation coded in Dear ImGui.	Editor research written. Editor wireframe complete. Editor foundation coded. Automated unit tests passing.
2. Business	SD Creations formalized as an LLC. IP and licensing strategy researched and documented	LLC Formed. Written IP research summary. Beginning pursuit on IP efforts.
3. Learning resources	Public website with engine documentation created. initial ECS tutorials created.	Domain and site created. Finalized documentation released. Learning resources created and stored.
4. Community	Active public presence documenting the build, growing an audience, and finding potential users.	Create basic website with About Us and Dev Logs. Create social media with posts pointing users to website.

By May 2027, ImagoEngine will have a working ECS core library, physics engine, asset renderer, and the foundation of a visual editor. Similarly, SD creations will be a formally

structure company with a public face, social media presence, and growing community. The project will be read to enter its serialization, editor polish, and scripting layer phase and SD creations will be ready to begin finding potential users, release our documentation and learning resources, and pursue intellectual property efforts.